Maxwell Kroft

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Education

Ph.D. **University of Wisconsin-Madison**, Astronomy — 2023 – · · · ·
- Advisor: Thomas Beatty

M.S. **University of Wisconsin-Madison**, Astronomy — 2023 – 2025

B.S. **University of California, Los Angeles**, Astrophysics — 2018 – 2022
- Graduated *Cum Laude*

Research Experience

University of Wisconsin-Madison, Department of Astronomy

Graduate Research Assistant — 2023 – · · · ·
- Advisor: Thomas Beatty

University of California, Los Angeles, Department of Physics & Astronomy

Post-Baccalaureate Lab Assistant — 2022 – 2023
- Advisors: Andrea Ghez, Shoko Sakai

Undergraduate Research Assistant — 2020 – 2022
- Advisors: Andrea Ghez, Shoko Sakai

Teaching Experience

University of Wisconsin-Madison, Department of Astronomy

Graduate Teaching Assistant, Astronomy 103 — Spring 2024

UCLA Math Circle

Assistant Teacher, 1st & 2nd grade class — 2018 – 2019

Publications

1st Author

GJ 523b is a Massive, 170 Myr-old Mega-Earth on a Likely Polar Orbit — Submitted to AJ Mar 2026
Maxwell Kroft, Thomas Beatty, Joseph Salzer, et al.
10.48550/arXiv.2603.24682

A Pair of Dynamically Interacting Sub-Neptunes Around TOI-6054 — Sep 2025
Maxwell Kroft, Thomas Beatty, Ian Crossfield, et al.
10.3847/1538-3881/adee24

2nd or 3rd Author

A New Astrometric Reference Frame for Measuring Precession of Stellar Orbits Around the Supermassive Black Hole at the Galactic Center: Leveraging the Strengths of W. M. Keck Observatory Adaptive Optics, Hubble Space Telescope, and GAIA — Submitted to ApJ Oct 2026
Shoko Sakai, ***Maxwell Kroft***, Asad Ukani et al.

Contributing Author

Stellar Orbits at the Galactic Center: New Estimates of the Black Hole Properties and the First Measurement of Orbital Precession with Keck Observatory — Submitted to ApJ May 2026
Andrea Ghez, Gregory Martinez, Zoë Haggard, et al. incl. ***Maxwell Kroft***

TOI-1355 b and TOI-1414 b: Two Giant Objects on Misaligned Orbits Around Hot Stars — Accepted by A&A Oct 2026
Zuzana Balkóová, Jiri Zak, M. Starka, et al. incl. ***Maxwell Kroft***

Discovery and Characterization of the TOI-4468 Planetary System: A Transiting Hot Jupiter With a Lone Nearby Outer Companion — Oct 2026
Joseph Livesey, Benjamin Hord, Juliette Becker, et al. incl. ***Maxwell Kroft***
10.3847/1538-3881/aeaee4

The TESS Grand Unified Hot Jupiter Survey. III. Thirty More Giant Planets. — Sep 2025
Samuel Yee, Joshua Winn, Joel Hartman, et al. incl. ***Maxwell Kroft***
10.3847/1538-4365/aded0d

OrCAS: Origins, Compositions, and Atmospheres of Sub-Neptunes. I. Survey Definition — Feb 2025
Ian Crossfield, Alex Polanski, Paul Robertson, et al. incl. ***Maxwell Kroft***
10.3847/1538-3881/ad9aa6

Software

Alfred: Awesome Library For Robust Exoplanet Detection
Github, Docs
Maxwell Kroft 2026, 10.3847/2515-5172/ae9165

Observing Experience

PI

NEID Confirmation of Bright Exoplanets 2027A — 2027A
NEID, WIYN Telescope, 50.03 hours awarded

Using NEID Spectra to Mitigate the Transit Light Source Effect — 2026B
NEID, WIYN Telescope, 1.33 hours awarded

Validating the Masses of Mega-Earths — 2026B
NEID, WIYN Telescope, 18 hours awarded

NEID Confirmation of Bright Exoplanets 2026B — 2026B
NEID, WIYN Telescope, 25.23 hours awarded

FASHION (Faint Accelerating Stellar Hosts In Our Neighborhood) — 2026B
NEID, WIYN Telescope, 6.8 hours awarded

Starspot Characterization with High Resolution Optical Spectra (GHOST) — 2026A
GHOST, Gemini South, 2.75 hours awarded.

Starspot Characterization with High Resolution Optical Spectra (LCO) — 2026A
Sinistro, Las Cumbres Observatory, 20.0 hours awarded.

NEID Confirmation of Bright Exoplanets 2026A — 2026A
NEID, WIYN Telescope, 47.07 hours awarded

FASHION (Faint Accelerating Stellar Hosts In Our Neighborhood) 2026A
NEID, WIYN Telescope, 9.46 hours awarded

NEID Confirmation of Bright Exoplanets 2025B 2025B
NEID, WIYN Telescope, 25.58 hours awarded

Refining the Masses of the TOI-1470 System 2025B
NEID, WIYN Telescope, 20.65 hours awarded, CO-PI Collette Sarver

CO-I

Hunting for Massive Black Holes with Gaia DR-4 and Follow-Up Spectroscopy 2027A
NEID, WIYN Telescope, 35 hours awarded, PIs Thomas Beatty and Sukanya Chakrabarti

RV Confirmation of NIR Bright Sub-Neptunes 2025A
NEID, WIYN Telescope, 45.52 hours awarded, PI Thomas Beatty

RV Confirmation of Candidate Hycean Planet GJ 523b 2025A
NEID, WIYN Telescope, 12.9 hours awarded, PIs Thomas Beatty and Juliette Becker

Confirmation of Bright Exoplanet Candidates Using NEID 2024B
NEID, WIYN Telescope, 39.83 hours awarded, PI Thomas Beatty

RV Confirmation of NIR Bright Sub-Neptunes 2024A
NEID, WIYN Telescope, 16.82 hours awarded, PI Thomas Beatty

Awards and Honors

NASA Wisconsin Space Grant Consortium's Graduate & Professional Research Fellowship, 2026
Blair Savage Astronomy Travel Fund, Department of Astronomy, University of Wisconsin-Madison, 2026
NSF Graduate Research Fellowship Program Honorable Mention, 2025
Chambliss Astronomy Achievement Student Award, AAS 244, 2024

Scientific Presentations

Talks

Using the Relationship Between GJ 523b and its Host Star to Inform its Evolutionary History and Composition May 2026
Astrobiology Science Conference 2026, Madison, WI

Searching for Hycean Worlds: GJ 523b Nov 2025
Great Lakes Exoplanet Area Meeting 2025, Madison, WI

Posters

Using the Relationship Between GJ 523b and its Host Star to Inform its Evolutionary History and Composition July 2026
Exoplanets 6, Porto, Portugal

TOI-6054: A Highly Irradiated, Resonant Pair of Sub-Neptunes Sep 2024
WiCOR Kickoff Reception, Madison, WI

A Pair of Hot Sub-Neptunes Around an F-Type Sub-Giant Jun 2024
AAS 244, Madison, WI

Outreach

Public Talks

Top 10 Least Habitable Planets in the Known Universe — May 2026
Badgers on Tap, Madison, WI

How do we *actually* know things about exoplanets? — April 2026
Invited talk at the Madison Astronomical Society, Madison, WI

How do we know stuff about stars and exoplanets?
Invited Badger Talk at Kirkland Crossings Senior Living Center, Pewaukee, WI — Jan 2025
Universe in the Park Program
- Kohler-Andrae State Park, Sheboygan, WI — Aug 2025
- Interstate State Park, St. Croix Falls, WI — Jul 2025
- Pattison State Park, Superior, WI — Aug 2024
- Mirror Lake State Park, Baraboo, WI — Jul 2024

How do we actually find exoplanets? — Dec 2024
Madison Astronomy on Tap, Madison, WI

Professional Development

Great Lakes Exoplanet Area Meeting 2025 Science Organizing Committee Member — Nov 2025
UW-Madison, Madison, WI

Summer Training in Astrobiology Research Bootcamp — Aug 2025
WiCOR, UW–Madison, Madison, WI

Code/Astro — Aug 2025
CIERA, Northwestern University, Evanston, IL

Employment

Private Math Tutor — 2019 – 2022

YMCA Camp Minikani, Hubertus, WI
Adventure Challenge Director — 2021
Summer Camp Counselor — 2017 – 2021

Technical Skills

Languages: Python, C++, C, Java

Software: Unix, Git, LaTeX, Mathematica, Excel